



Using the fx-991CW for statistics

Task A: Finding the mean from a frequency table

1) This frequency table shows the number of goals scored in each match by all the teams in a club over one football season.

Goals scored	Frequency
0	19
1	34
2	21
3	15
4	11

What was the mean number of goals scored?

1.65 goals (which rounds to 2)

2) This frequency table shows the star rating received by a restaurant over its opening weekend.

Star rating	Frequency
1	12
2	18
3	21
4	38
5	45

What was the mean star rating?

3.64... stars (which rounds to 4)

3) The frequency table shows the number of bedrooms in different houses within an area of a town.

Bedrooms	Frequency
1	20
2	93
3	54
4	33

What is the mean number of bedrooms?

2.5 bedrooms (which rounds to 3)

4) The time it takes for a sample of pupils to travel to school is collected in this grouped frequency table.

Time taken to travel to school (in minutes)	Frequency (<i>f</i>)
$0 < t \leq 4$	28
$4 < t \leq 10$	23
$10 < t \leq 16$	70
$16 < t \leq 26$	52
$26 < t \leq 40$	30
$40 < t \leq 60$	13

What is the estimated mean travel time (to the nearest minute)?

18 minutes

Task B: Analysis

1) This frequency table shows the number of goals scored in each match by all the teams in a club over one football season.

Goals scored	Frequency
0	19
1	34
2	21
3	15
4	11

a) Calculate the median.

1 goal

b) Calculate the interquartile range.

$3 - 1 = 2$ goals

c) Would the club prefer to report the mean or median as the average number of goals scored? Why?

Mean = 1.65 goals. Median = 1 goal.

The club would prefer to report the mean as it rounds to 2 goals making the club's performance look better.

2) This frequency table shows the star rating received by a restaurant over its opening weekend.

Star rating	Frequency
1	12
2	18
3	21
4	38
5	45

a) Calculate the median.

4 stars

b) Calculate the interquartile range.

$5 - 3 = 2$ stars

c) Would the restaurant prefer to report the mean or median as the average rating? Why?

Mean = 3.64... stars. Median = 4 stars.

The restaurant could report either but the median does not need to be rounded.



3) The frequency table shows the number of bedrooms in different houses within an area of a town.

Bedrooms	Frequency
1	20
2	93
3	54
4	33

A developer wishes to build more expensive houses (3+ bedrooms) but the local council want cheaper housing. The council argue that there are already enough houses with 3 or more bedrooms.

Are the council correct? Use the data to support your decision.

Mean = 2.5 bedrooms. Median = 2 bedrooms.

Interquartile range = 3 - 2 = 1

E.g.
 If the council uses the mean, then they can round and state that the average number of bedrooms in houses in this area is 3. Therefore there are plenty of houses with 3 (or more) bedrooms in.

4) The time it takes for a sample of pupils to travel to school is collected in this grouped frequency table.

Time taken to travel to school (in minutes)	Frequency (f)
$0 < t \leq 4$	28
$4 < t \leq 10$	23
$10 < t \leq 16$	70
$16 < t \leq 26$	52
$26 < t \leq 40$	30
$40 < t \leq 60$	13

It takes 17 mins for Jacob to travel to school by bus.

Jacob claims this takes him longer than the average pupil so he should be taken to school by car every day instead.

Do you agree or disagree with Jacob? Use evidence to support your answer.

Mean $\approx$ 18 minutes. Median = 13 minutes.

If Jacob uses the median time then he is correct.

However, Jacob's family could argue he is wrong by using the mean time.